



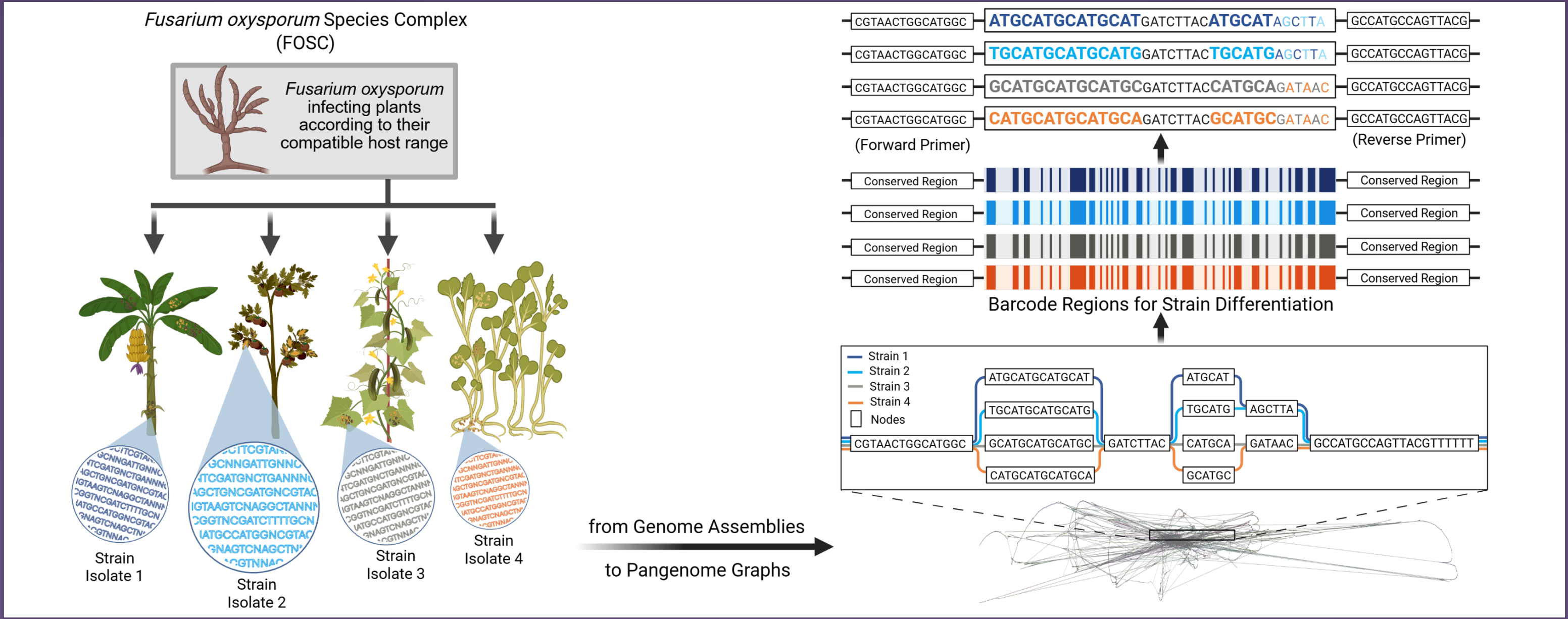
Decoding *Fusarium oxysporum* pathogenicity and variability with pangenome graphs

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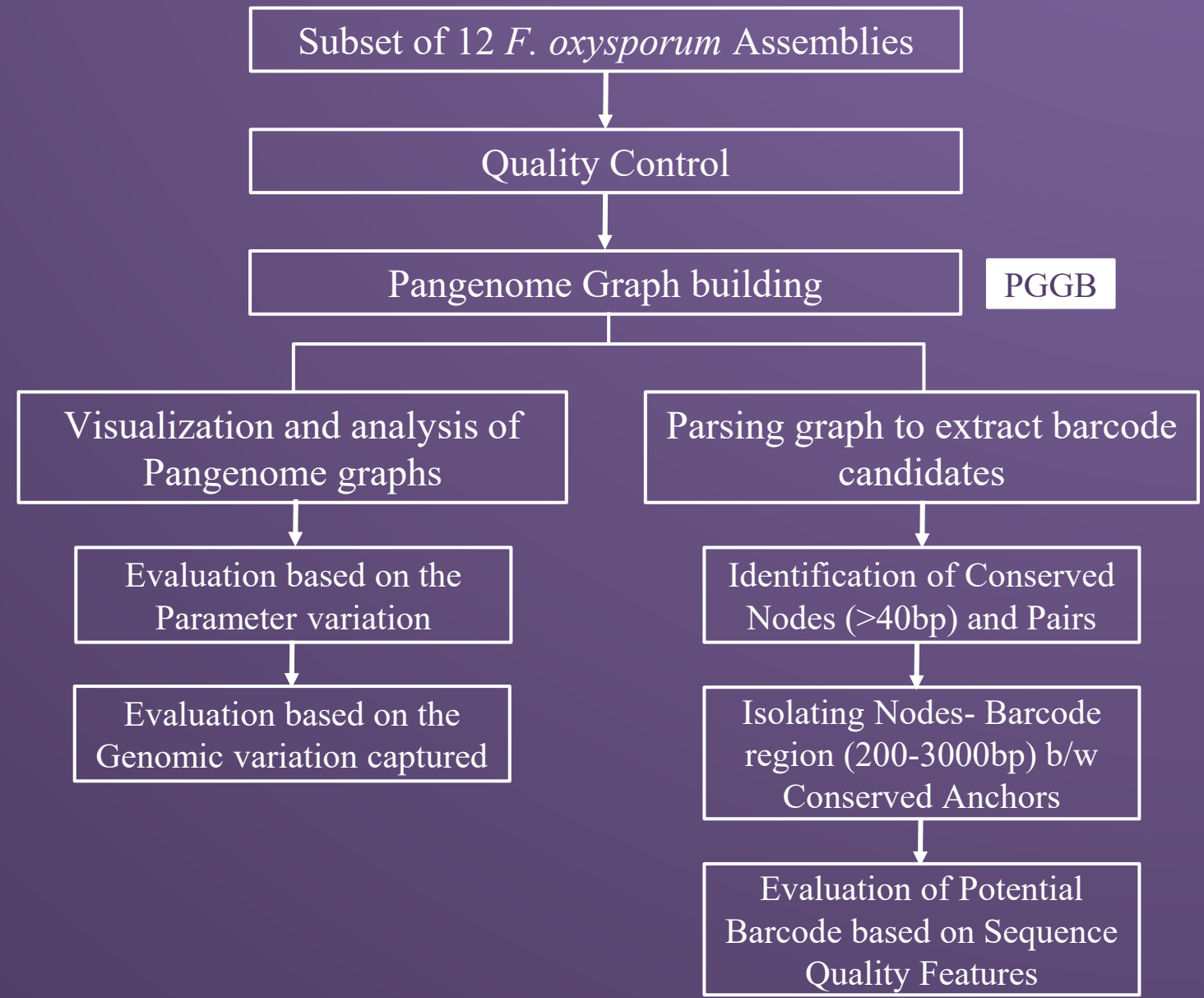
Overview



Introduction

Fusarium oxysporum is a diverse species complex containing both harmless endophytes and devastating plant pathogens. Differentiating pathogenic from asymptomatic strains is essential but challenging, as standard DNA markers often lack resolution. We aim to develop long-read DNA barcodes that reliably identify high-risk pathogenic lineages. Using pangenome graphs, which capture the full genetic diversity across strains, we build a comprehensive framework for strain comparison and classification. Our benchmark pangenome will support the design of diagnostic barcodes and enable an open-source tool for detecting strains based on key traits like host specificity and pathogenicity, advancing molecular diagnostics and biosecurity.

Methodology



Results

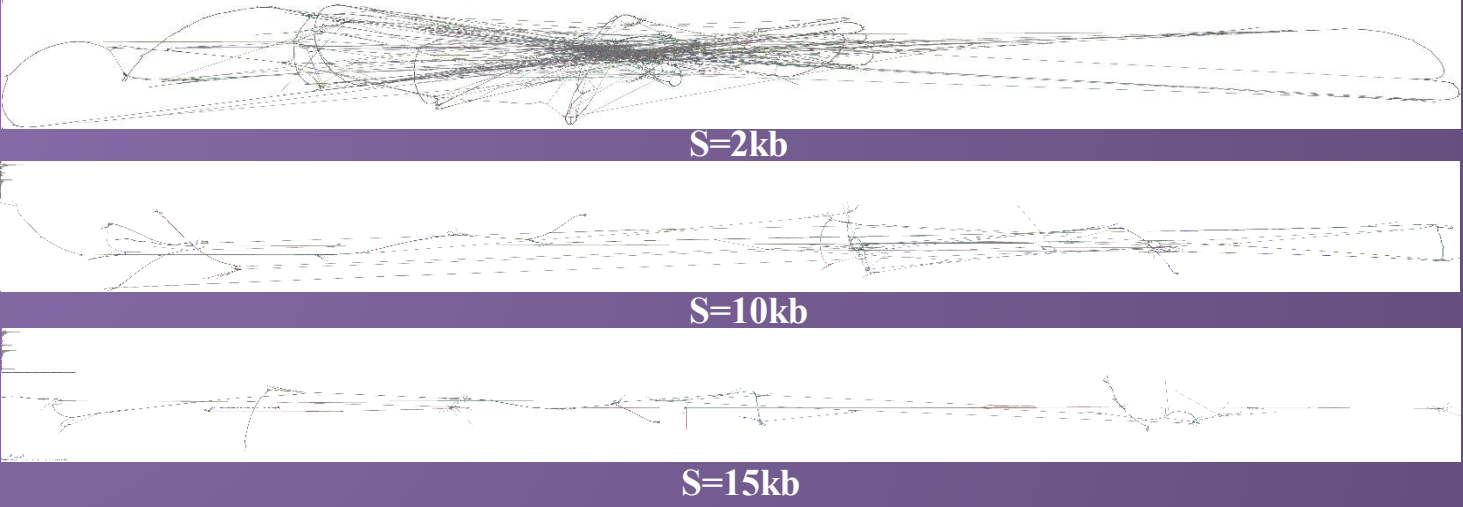


Fig 1: Analyzing and differentiating pangenome graphs based on the Segment Length parameter variation

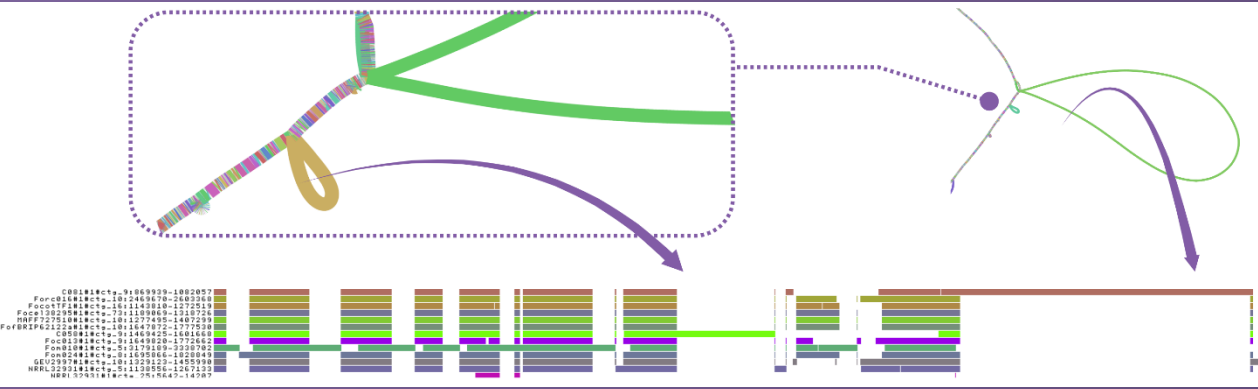


Fig 2: Observing Conserved regions (multi-colored), Insertions (green) and large Starship sequences (brown)

Conserved Nodes	Node Pairs	Barcode Candidates
156,634	1,879,507	80,810

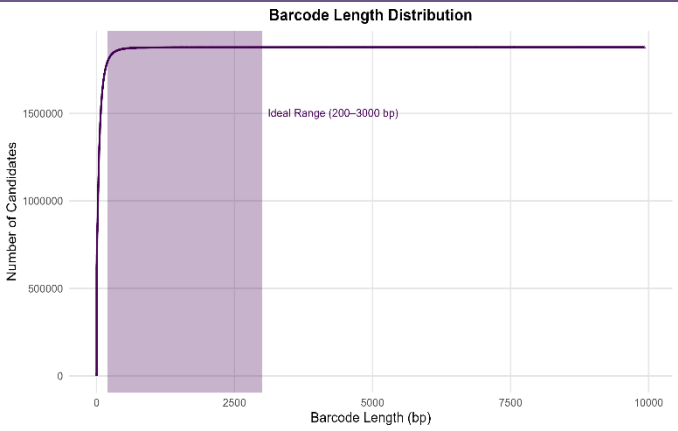
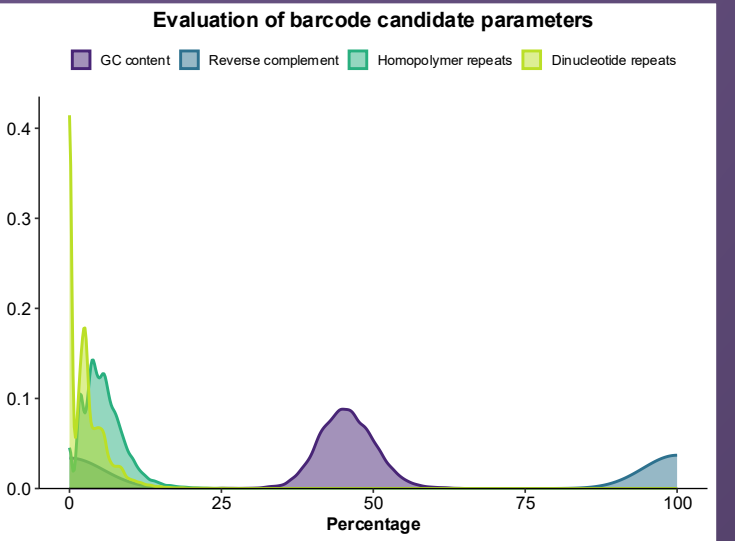


Fig 3: Filtering potential barcodes within the ideal amplicon size range and their characteristics



Conclusion

We identified over 80,000 candidate barcode regions in a pangenome graph, filtered by sequence length and quality features. This highlights the potential of graph-based approaches for discovering variability suitable for diagnostic or genotyping purposes.

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